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United States Patent	12397977
Kind Code	B2
Date of Patent	August 26, 2025
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Dual-purpose container closure

Abstract

A closure for a beverage container has a body having an attachment interface to engage to a top of a beverage container, a sealed, elongated enclosure attached to an underside of the body, and a single access element adapted to selectively open and close either a first opening passing through the body into the sealed, elongated enclosure or a second opening passing through the body directly to an inner volume of the beverage container.

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Family ID:	1000008778780
Appl. No.:	18/149499
Filed:	January 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230150738 A1	May. 18, 2023

Related U.S. Application Data

continuation-in-part parent-doc US 17196625 20210309 US 11542077 child-doc US 18149499
division parent-doc US 16797665 20200221 US 10974881 20210413 child-doc US 17196625

Publication Classification

Int. Cl.: **B65D51/28** (20060101); **A24F13/12** (20060101); **A24F15/01** (20200101); **B65D25/04** (20060101); **B65D43/02** (20060101); **B65D47/06** (20060101); **B65D51/16** (20060101)

U.S. Cl.:

CPC **B65D51/28** (20130101); **A24F13/12** (20130101); **A24F15/01** (20200101); **B65D25/04** (20130101); **B65D43/0231** (20130101); **B65D47/065** (20130101); **B65D51/1633** (20130101);

Field of Classification Search

CPC: A24F (7/00); A24F (1/14); A24F (9/14); A24F (15/01); A24F (13/12); A24F (1/30); A24F (47/008); B65D (51/28); B65D (25/04); B65D (43/0231); B65D (47/065); B65D (51/1633); B65D (25/20); B65D (51/2807); A61M (2205/59); A45F (3/16); A62B (18/086); A47G (19/22)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present invention is a continuation-in-part of co-pending application Ser. No. 17/196,625, filed Mar. 9, 2021, which is a divisional application of Ser. No. 16/797,665 filed Feb. 21, 2020, now issued as U.S. Pat. No. 10,974,881 on Apr. 13, 2021. All disclosure of the parent application is incorporated at least by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention is in the technical area of apparatus and methods for smoking tobacco and other materials, and pertains more particularly to a container closure that enables a user to access a vapor pen enclosed in the container, or to ingest liquid from the container as well.

2. Description of Related Art

(2) There exist in the conventional art a considerable variety of beverage containers, such as water and coffee containers, adapted for users to fill and refill, and to ingest liquid from the container as desired. There exist in the art as well a considerable variety of vapor pens and the like for users to draw flavored vapor and the like as desired. What is needed is a closure for existing beverage containers, the enclosure having apparatus with alternative draw elements, that allows drawing liquid, such as coffee or water from the container by one draw element, and allows drawing from an enclosed vapor-producing apparatus through a separate draw element.

BRIEF SUMMARY OF THE INVENTION

(3) In an embodiment of the invention a closure for a beverage container is provided, comprising a body having an attachment interface to engage to a top of a beverage container, a sealed, elongated enclosure attached to an underside of the body, and a single access element adapted to selectively open and close either a first opening passing through the body into the sealed, elongated enclosure

or a second opening passing through the body directly to an inner volume of the beverage container. In one embodiment the attachment interface of the body is a male thread matching a female thread at the top of the beverage container. Also, in one embodiment the single access element is an elongated element having a cylindrical portion at a first end and an extended portion from the cylindrical portion with a passage through a length of the element, the cylindrical portion pivoted at a center point of the cylinder between opposite sides of an elongated well on an upper region of the body, in a manner that with the access element in a first position lying in the elongated well with the cylindrical portion engaging a cylindrically shaped depression the access element seals a vent into both the beverage container and the sealed enclosure, with the access element rotated to a second position aligns the passage through the length of the access element with an opening through the body into the sealed enclosure and with the access element rotated to a third position aligns the passage through the length with an opening through the body into an inner volume of the beverage container.

(4) In one embodiment the first opening passing through the body into the sealed elongated enclosure ends at a first interface under the body with a first tube engaged reaching proximate the lower end of the sealed elongated enclosure, and the second opening passing through the body ends at a second interface with a second tube engaged reaching proximate a bottom of the beverage container. Also, in one embodiment the first opening passing through the body into the sealed elongated enclosure ends at a first interface under the body with a flexible connector engaged to the first interface and to a vapor-producing apparatus and the second opening passing through the body ends at a second interface with a tube engaged reaching proximate a bottom of the beverage container.

(5) In one embodiment the attachment interface of the body is a female thread matching a male thread at the top of the beverage container. In one embodiment the attachment interface of the body is a male thread matching a female thread at the top of the beverage container. And in one embodiment the access element has three spherical depressions on one side on a radial ring and a spring-loaded ball on a surface of an inside wall of the rectangular well, such that the ball engages the depressions as the access element rotates and positions the access element accurately at the first, the second and the third position.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) FIG. 1 is an elevation view of a container closure engaged to a container in an embodiment of the invention.
- (2) FIG. 2 illustrates the container closure of FIG. 1 removed from the container and showing an inner sealed enclosure.
- (3) FIG. 3 is a partial section view through the closure of FIG. 1.
- (4) FIG. 4 is an elevation view, in section, of the inner sealed enclosure of FIG. 2.
- (5) FIG. 5 is a section through the container closure of FIG. 1.
- (6) FIG. 6 is a partially sectioned view of an adapter shown in FIG. 4.
- (7) FIG. 7 is a perspective view of a closure for a container in an alternative embodiment of the invention.
- (8) FIG. 8 is a cross section of the closure of FIG. 7.
- (9) FIG. 9 is a perspective view of a container closure having a single access element for selectively accessing two inner regions of a beverage container.
- (10) FIG. 10 is an elevation cross section of the container closure of FIG. 9 with the access element rotated to a closed position.
- (11) FIG. 11 is the section of FIG. 10 with the access element rotated to access a sealed enclosure.

- (12) FIG. 12 is the section view of FIG. 10 with the access element rotated to access an inner volume of the beverage container.
- (13) FIG. 13 is the section view of FIG. 10 with the access element rotated closed and with a vapor producing apparatus in the sealed enclosure.
- (14) FIG. 14 is the section view of FIG. 13 with the access element rotated to access the vapor producing apparatus in the sealed enclosure.
- (15) FIG. 15 is the section view of FIG. 13 with the access element rotated to access the inner volume of the beverage container.
- (16) FIG. 16A is an elevation view of the access element with depressions for positioning.
- (17) FIG. 16B illustrates a spring-loaded ball positioning the access element.

DETAILED DESCRIPTION OF THE INVENTION

- (18) FIG. 1 is an elevation view of a container closure **101** engaged to a container **102** in an embodiment of the invention. The container may be any one of a variety of conventional beverage containers known in the art, such as, for example, an aluminum coffee container like many provided by coffee enterprises. Closure **101** in this embodiment of the invention comprises a body **103**, which in this example has an internal female thread to engage a male threaded top of container **102**. In alternative embodiment the closure may have a body with a male thread to engage a female threaded top of a container. In other embodiments the closure may have an interface to engage containers of other sorts, such as even paper cups and the like.
- (19) In the example shown in FIG. 1 closure **101** has two access wells **104** and **105**. Access well **104** has a pivoted access element **106**, that tipped up opens a path from a draw tip down through the closure into a second sealed enclosure within the outer enclosure of container **102**. In this example a vapor-producing apparatus such as a vape pen is connected within the second sealed enclosure through access element **106**, enabling a user to tip up element **106**, and to draw vapor from the vapor-producing apparatus.
- (20) Access well **105** has a pivoted access element **107**, similar in design and function to access element **106**, that connects to a tubing under the closure, the tubing extending downward into any liquid, such as water or coffee, that may be carried in container **102**. A user may tip up access element **107** and draw liquid through the tube and the access element as one would using a straw. There is a vent hole not shown that serves to equalize pressure. Finally, a handle **108** is provided on the closure to carry the closure and any added container.
- (21) FIG. 2 illustrates closure **101** removed from container **102**, showing the second inner sealed enclosure **201**, that may enclose, for example, a vapor-producing apparatus, and also a tubing **202** that extends downward, and may extend into any liquid that may be carried in a container like container **102**. It should be noted that enclosure **201** is completely enclosed and sealed from liquid that may be in container **102**, while tube **202** is open to the liquid, and extends downward into the liquid, which may be water, coffee or any other beverage. Tube **202** may be fashioned of a suitable polymer material.
- (22) FIG. 3 is a partial section view through closure **101**, taken along a centerline of well **104** and access element **106**. It may be seen in this section that well **104** is a rectangular enclosure with an open top. Access element **106** is an elongated unit of a width to fit into the width of well **104**. There is a passage **303** through the length of element **106**, with an opening **304a** on one end and an opening **304b** on the other. The end of element **106** with opening **304b** has a spherical shape and opening **304b** opens through this spherical shape. Access element **106** is pivoted across well **104** along an axis at the center of the spherical shape, such that a user may use tip **301** to rotate the access element around the pivot access to a stop point where opening **304b** aligns with a vertical passage **309** through closure **101**.
- (23) A seal element **305** locates in a passage **308** and has a concave spherical shape facing upward into well **104**, and the spherical shape **302** of access element **106** rests in and rotates against seal element **305**. At the rotation stop point that opening **304b** aligns with vertical passage **309**, a user

may place lips on the end of element **106** with opening **304b**, and may draw on the volume into which vertical passage **308** opens below closure **101**.

(24) In addition to access to passage **309** through the passage **303** through access element **106**, there is a small vent hole **311** vertically through seal element **305**, and this hole communicates with a passage **312** into the volume below closure **101**. A small tip **310** on the underside of access element **106** closes hole **311** when the access element is rotated to be fully within well **104**. Passage **309** leads to and through a cylindrical nib **315**.

(25) A male threaded portion **314** on a lower portion of closure **101** provides for engagement of an elongated enclosure **201** (see FIG. 2) by a female thread on the top of enclosure **201**. There is also a gasket **307** against which enclosure **201** seats when fully engaged by threaded portion **314**. Internal female threads **313** are to engage with external male threads at the top of beverage container **102**, and a gasket **306** engages a top rim of the beverage container to provide a seal.

(26) FIG. 4 is an elevation view, in section, of inner sealed enclosure **201** engaged by internal threads to threads **314** of closure **101**. Sealed enclosure **201** fully engaged by threads **314** urges against a gasket **307**, also seen in FIG. 3, by which the inner volume of enclosure **201** is isolated from the volume within beverage container **102**. In one embodiment of the invention an adapter **401**, molder of a flexible polymer, has a circular opening on one end to connect to nib **315**, and opening on the opposite end to connect to a vapor-producing apparatus **402**, which may be a conventional vape pen of one of several types. When access element **106** is fully rotated to a stop to align opening **304b** with passage **309** a user may draw on opening **304a** to ingest vapor from vapor-producing apparatus **402**. Passage **312** provides a vent to atmosphere for the volume in enclosure **201** outside the vapor-producing apparatus.

(27) FIG. 5 is a section through closure **101** along a centerline of access element **107** in well **105**. Access element **107** differs from access element **106** by the placement of a tip **507** in a position to close a vent opening **508** through a bottom surface of well **105** into the inner volume of beverage container **102**. Vent opening **508** has a seal element **509** to secure a good seal when access element **107** is rotated down to close opening **508** by tip **507**.

(28) A seal element **505** similar to seal element **307** of FIG. 3 surrounds an opening to passage **506** in a bottom surface of well **105**. Access element **107** has a spherically-shaped end **502** the matches a spherical shape of seal element **505** such that access element **107** may be rotated by tip **501** to align opening **504b** with passage **506**. Passage **506** terminates at a nib **510** to which tubing **202** connects, by virtue of tubing **202** being formed of a flexible polymer material. In an alternative embodiment tubing **202** may connect to nib **510** by a flexible tubing adapter piece. With access element **107** fully rotated to a stop and opening **504b** aligned with passage **506** a user may draw on opening **504a** to bring liquid up from beverage container **102**, through passage **506** and passage **503**, into the user's mouth.

(29) It will be apparent to the skilled person that the closure **101** as described above, in concert with inner enclosure **201**, adapter **401** and vapor apparatus **402**, and tubing **202** may selectively open and close the two separate access elements and access either vapor from vapor apparatus **402** or sips of beverage from beverage container **102**.

(30) FIG. 6 is a partially sectioned view of adapter **401** of FIG. 4, adapted to connect nib **315** and vapor apparatus **402**. Region **402** is circular with circular internal passage **404**, which is pushed onto nib **315** to connect to passage **309**. A wider portion **408** has differently shaped internal regions **405**, **406** and **407** to engage tip ends of different vapor-producing apparatus. In the example illustrated by FIG. 4 the engagement is by region **407**, as the vapor-producing apparatus of FIG. 4 has a wider and flat aspect. In one embodiment adapter **401** is molded to accommodate and connect to three of the more preferred conventional vapor-producing apparatus.

(31) In the figures and description above closure **101** is illustrated and described as having internal threads to match external threads on an upper extremity of a beverage container. In an alternative embodiment the beverage container may have an internal threaded upper region, and closure may

have an externally threaded region to engage the internal threads of the beverage container. In yet another embodiment the closure may be adapted to snap fit on the upper rim of a beverage container having a plain upper rim, like, for example a paper cup.

(32) FIG. 7 is a perspective view of a closure **701** adapted to snap onto a thin-walled beverage container **707**, such as a paper cup, for example. In this example the closure has an upper body **702** that is molded of a polymer material and is substantially rigid. A lower body **703** has a relatively thin wall to allow the lower body some flexibility to engage an upper rim of the beverage container **707**. A substantially rectangular well **704** is formed in the upper body, and a pivoted cover **705** engages inner walls of the well. A second well **709** is formed in the upper body and an access element **706** engages inner walls of well **709**. Access element **706** is similar to access elements **106** and **107** of FIG. 2, and has an opening **708** to a channel through the length of the access element.

(33) FIG. 8 is a cross section of closure **701** taken along a centerline that passes lengthwise through both pivoted cover **705** and access element **706**. The thin-wall lower body **703** is illustrated as snapping over an upper rim of paper cup **707**. In this embodiment there is a gasket seat **710** in a vertical opening through closure body **702** that presents an opening **711** that is closed by a shape of pivoted cover **705** with the pivoted cover in a downward horizontal closed position. When pivoted cover **705** is pivoted upward opening **711** is exposed, and provides an interface for a user to place the lips at the edge of the closure, tilt the beverage container and drink from the container just as a user would an open container. Opening the pivoted cover also opens a vent hole **713** that is closed by tip **712** with the pivoted cover in the closed position.

(34) Access element **706** is pivoted within well **709** just as access elements **106** and **107** in wells **104** and **105**, and access element **706** has a spherically-shaped end **716** that mates with a spherical shape of a gasket seat **717**, such that access element **706** may be rotated to align opening **715** with opening **718**, such that channel **720** is open through opening **718**, opening **715**, channel **714** and opening **708** to the outside.

(35) In this example male threads **719** are the same as threads **314** shown in FIG. 3, such that an inner enclosure **201** may be engaged by female threads to male threads **719** just as illustrated in FIG. 4, to bear on a gasket **724**. Adapter **401** of FIG. 4 may be engaged to nib **721** and to a vapor-producing apparatus **402**, just as illustrated in FIG. 4.

(36) Closure **701** is thus a dual-access container closure just as is closure **101**. A user may manipulate pivoted cover **705** to drink from the paper cup, and may manipulate access element **706** to draw vapor from a vapor-producing apparatus such as apparatus **402** and others, adapted to the closure by an adapter such as that illustrated in FIG. 6.

(37) A skilled person will understand that the embodiments illustrated in FIGS. 2 and 3 may be adapted to snap onto a paper cup, such as beverage container **707** of FIGS. 7 and 8, as shown in FIG. 8, and also that the embodiment shown in FIG. 8 might also be adapted to have a male or a female thread to engage threads of a beverage container, as described above for the embodiment illustrated by FIGS. 2 and 3.

(38) In yet another embodiment of the invention, referring again to FIG. 2 and to FIG. 4, the adapter **401** and the vapor-producing apparatus may be removed from inside the sealed enclosure **201**, and a tubing similar to tubing **202** may be connected to nib **315** to extend down into the interior of the sealed enclosure. In this arrangement one beverage may be introduced into container **102** outside the sealed enclosure **201**, and a second, different beverage may be introduced into sealed enclosure **201**. In this arrangement the two beverages are completely separated, and do not mix, and a user may draw one beverage from the container outside the sealed enclosure, and a different beverage from inside the sealed enclosure within the beverage container. As a simple example, one may have water in one place and vodka in another and may manipulate the access elements to imbibe one or the other.

(39) In the embodiments described above a container closure provides two separate enclosures in the beverage container and two access mechanisms, one for each of the separate enclosures. See

FIGS. 1 and 2 and descriptions in the specification above. In yet another embodiment of the invention a single access container closure is provided in a manner that a user may access the two separate enclosures with a single access mechanism in a single access well.

(40) FIG. 9 illustrates a closure **901** with a threaded body **902** for closing a beverage container. There is a sealed enclosure **905** analogous to sealed enclosure **201** of FIG. 2. There is in addition a tube **906** that communicates directly with the internal volume of the beverage container. Body **902** has a single rectangular access well **903** with one tip-up access element **904**, which is pivoted on the sides of the well at point **907**. The access well and access element are devised such that a user may selectively access either the internal volume of sealed enclosure **905** or the internal volume of the beverage container.

(41) FIG. 10 is a section view of container closure **901** taken on a vertical section plane that passes through the center of body **902**, access well **903** and element **904** lengthwise. Body **902** has a female thread **1001** to engage male thread on a beverage container. In this example there is a sealed enclosure **905** that has a female thread on an upper end that engages a male thread on an extension **1002** under body **902**. A gasket **1003** at the upper end of extension **1002** seals the inner volume of enclosure **905**, and in this example a tube **1015** engages an extension **1004** and opens into the volume of enclosure **905**. A tube **906** engages a bore in an extension **1006** and opens into the main volume of the beverage container outside of sealed enclosure **905**.

(42) A tip-up access element **904** has a cylindrical end **1008**, an extension **1010** and a tab **1011** that is useful for tipping the access element. Cylindrical end **1008** is pivoted across sides of a rectangular well **903** and rests in a cylindrical seat in a bottom of the well, the seat lined by a seal **1009** such that the outside diameter surface of the cylindrical portion of the tip-up element remains sealed to body **902** as the access element is rotated.

(43) Access element **904** has a lengthwise through channel **1012** that extends the full length of the access element. There is a tubular channel **1005** through body **902** and gasket **1009** that leads to tube **1015**, and another tubular channel **1007** through body **902** and gasket **1009** that leads to tube **906**. A bore **1013** leads from a bottom of access well **903** to the inner volume of the beverage container, with a seal **1014** at the upper end. Tip-up element **904** has a side tip **1016** (FIG. 11) that closes bore **1013** when the tip-up access element **904** is closed in the well as shown in FIG. 10. A side passage of bore **1013** leads to the inner volume of enclosure **905**, but is not on the section plane so as to not intersect another bore or passage.

(44) FIG. 11 illustrates the container closure of FIG. 10 with tip-up element **904** rotated such that through channel **1012** aligns with tubular channel **1005** which communicates with tube **1015**. Note that tubular channel **1007** is closed with the access element thus rotated. In this position a user may imbibe a liquid in the internal volume of sealed enclosure **905**.

(45) FIG. 12 illustrates the container closure of FIG. 10 with tip-up element **904** rotated such that through channel **1012** aligns with tubular bore **1007** which communicates with the inner volume of the beverage container. In this position a user may imbibe a beverage in the inner volume of the beverage container. Given the description of FIGS. 11 and 12 it will be apparent that in this example a user may have two different beverages and access each by choice.

(46) FIG. 13 illustrates the closure of FIG. 10 with tube **1015** removed and a vape pen **402** engaged with a flexible adapter **401** which in turn engages extension **1004** from the underside of the body, all within sealed enclosure **905**. Description of adapter **401** and vape pens is provided above as well. In the example of FIG. 13 access element **904** is shown in the closed position in well **903** with channel **1013** sealed.

(47) FIG. 14 illustrates the closure of FIG. 13 with access element **904** rotated to a position to align channel **1012** with tubular channel **1005** enabling the user to draw on vape pen **402**. In this position channel **1013** is open venting the inner volume of sealed enclosure **905**.

(48) FIG. 15 illustrates the closure of FIG. 13 with the access element **904** rotated to position to align channel **1012** with tubular channel **1007** enabling the user to imbibe a liquid from the

beverage container. Again, channel **1013** is open at this position providing vent to the beverage container.

(49) In embodiments of the invention described here mechanisms are provided for aiding the user in positioning access element **904** rotationally. FIG. **16A** is a partial cutaway elevation view of access element **904** pivoted across sides of well **903** at a point P which is the center point of the spherical portion **1008** of the access element. In this example there are three small **1601**, **1602** and **1603** spherical indentions on a radius from the pivot point on one side of the access element to position the access element rotationally at three different positions, as are indicated in FIGS. **10-15**. FIG. **16B** illustrates a cutout section showing one indentation **1605** in access element **904** engaging a ball **1604** urged by a spring **106** in a bore in one side of well **903**. Ball **1605** engages each of the three indentions **1601**, **1602** and **1603** to accurately position the access element for alignment of channel **1202** with other channels.

(50) A skilled person will understand that the embodiments and examples presented and described above are all exemplary, and not limiting to the invention. Embodiments of the invention may incorporate features described in several variations. The scope of the invention is limited only by the claims.

Claims

1. A closure for a beverage container, comprising: a body having an attachment interface to engage to a top of a beverage container; a sealed, elongated enclosure attached to an underside of the body; and a single tip-up access element adapted to selectively open and close either a first opening passing through the body into the sealed, elongated enclosure or a second opening passing through the body directly to an inner volume of the beverage container, by a movement of the single tip-up access element.
2. The closure of claim 1 wherein the attachment interface of the body is a male thread matching a female thread at the top of the beverage container.
3. The closure of claim 1 wherein the single tip-up access element is an elongated element having a cylindrical portion at a first end and an extended portion from the cylindrical portion with a passage through a length of the element, the cylindrical portion pivoted at a center point of the cylinder between opposite sides of an elongated well on an upper region of the body, in a manner that with the single tip-up access element in a first position lying in the elongated well with the cylindrical portion engaging a cylindrically shaped depression the single tip-up access element seals a vent into both the beverage container and the sealed enclosure, with the single tip-up access element rotated to a second position aligns the passage through the length of the single tip-up access element with an opening through the body into the sealed enclosure and with the single tip-up access element rotated to a third position aligns the passage through the length with an opening through the body into an inner volume of the beverage container.
4. The closure of claim 1 wherein the first opening passing through the body into the sealed elongated enclosure ends at a first interface under the body with a first tube engaged reaching proximate the lower end of the sealed elongated enclosure, and the second opening passing through the body ends at a second interface with a second tube engaged reaching proximate a bottom of the beverage container.
5. The closure of claim 1 wherein the first opening passing through the body into the sealed elongated enclosure ends at a first interface under the body with a flexible connector engaged to the first interface and to a vapor-producing apparatus and the second opening passing through the body ends at a second interface with a tube engaged reaching proximate a bottom of the beverage container.
6. The closure of claim 1 wherein the attachment interface of the body is a female thread matching a male thread at the top of the beverage container.

7. The closure of claim 1 wherein the attachment interface of the body is a male female thread matching a female male thread at the top of the beverage container.

8. The closure of claim 3 wherein the single tip-up access element has three spherical depressions on one side on a radial ring and a spring-loaded ball on a surface of an inside wall of the rectangular well, such that the ball engages the depressions as the single tip-up access element rotates and positions the single tip-up access element accurately at the first, the second and the third position.
